

CMPG 122

User Interface Programming

Practical 6

Chapter 4: Making Decisions

Submission guidelines:

- Keep the HCI principles in mind when designing your GUI.
- Make sure to use “save all” when you save your project.
- Make sure that you provide the name, number and your student number for each program, e.g. **Prac1.1_Books_12345678**
- After completing all projects, combine ALL projects in the same folder.
- Zip only the ONE folder before you submit the practical on eFundi.

Practical 6

Create a Windows form application (.NET Framework) **called** *SurnameNamePrac6*.

Abahambe lifts is a South African lift company that provides people with lifts from one place to another. They have asked you to create an application that will book tickets and display confirmation of the booking to its clients. The objective of the application is to retrieve the pickup point and destination from the user and calculate transport fees.

The user must:

- Choose their pickup point
- Choose their destination
- Specify whether they want to book a return ticket or not.
- Specify how many tickets they want to buy

Below are tables showing what the respective costs are:

Pickup Point	Basic fee
Potch Ikageng	R250.00
Potch Promosa	R220.00
Potch Mohadin	R270.00

If the user does not choose a pickup point, the system can apply the Main Potchefstroom pickup point of R200. 00.

Destination	Additional fee
Johannesburg	R200.00
Pretoria	R250.00
Randfontein	R150.00
Durban	R600.00
Centurion	R250.00
Bloemfontein	R400.00
Capetown	R900.00

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Form design: Create a form in the same way you see it in the output example. The first 3 items in the listbox must be added using the properties window. The rest of the items must be added to the form using code.

Return trip Checkbox: Whenever the “Return trip” checkbox is clicked the user should be reminded that they will be booking a return trip and they will receive a 20% discount.

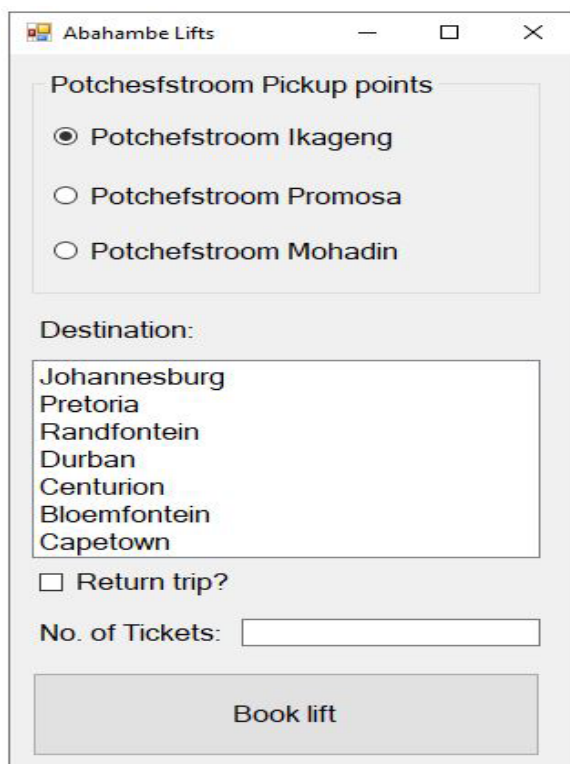
Book ticket Button: When the “Book ticket” button is clicked, the application must do the following:

The calculation may only go ahead with calculating the ticket fees if the number of tickets was successfully converted and no exceptions were thrown.

1. The basic fee must be set based on the pickup point (use the appropriate multiple alternative decision structure to do this).
2. The lift fare must be calculated based on the destination selected. The formula for lift fare is: **basic fee + additional fee**. Set the final lift fee using a switch statement.
3. The total lift fare should be updated and if there is more than one ticket purchased. The formula for total lift fare is : **lift fare * number of tickets**
4. Exceptions for listbox controls must be handled in the event a user does not select a destination.
5. If the user wants to book a return trip, a discount of 80% must be applied to their total lift fare. The formula is for discount is: **total fare * 0.8 * 2**.
6. The application must display a message with the final fees payable.

NB: Always remember that using comments is crucial, as is HCI compliance, the correct use of naming conventions for controls and exception handling.

See output examples below:



The screenshot shows a Windows application window titled "Abahambe Lifts". The form contains the following elements:

- Potchesfstrom Pickup points:** A group box containing three radio buttons:
 - ☒ Potchefstroom Ikageng
 - ☐ Potchefstroom Promosa
 - ☐ Potchefstroom Mohadin
- Destination:** A label followed by a list box containing the following items:
 - Johannesburg
 - Pretoria
 - Randfontein
 - Durban
 - Centurion
 - Bloemfontein
 - Capetown
- Return trip?:** A checkbox.
- No. of Tickets:** A text input field.
- Book lift:** A large button at the bottom of the form.

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Abahambe Lifts

Potchefstroom Pickup points

☐ Potchefstroom Ikageng

☒ Potchefstroom Promosa

☐ Potchefstroom Mohadin

Destination:

Johannesburg

Pretoria

Randfontein

Durban

Centurion

Bloemfontein

Capetown

☒ Return trip?

No. of Tickets:

Book lift

We will book you a return ticket, if this box is checked
You will receive a 20% discount

OK

Abahambe Lifts

Potchefstroom Pickup points

☐ Potchefstroom Ikageng

☒ Potchefstroom Promosa

☐ Potchefstroom Mohadin

Destination:

Johannesburg

Pretoria

Randfontein

Durban

Centurion

Bloemfontein

Capetown

☒ Return trip?

No. of Tickets:

Book lift

your 3 tickets have been booked to Randfontein is booked
Total payable is: R1 776,00

OK

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Abahambe Lifts

Potchefstroom Pickup points

☒ Potchefstroom Ikageng

☐ Potchefstroom Promosa

☐ Potchefstroom Mohadin

Destination:

Johannesburg
Pretoria
Randfontein
Durban
Centurion
Bloemfontein
Capetown

☐ Return trip?

No. of Tickets:

Book lift

Please choose a destination from the list

OK

Abahambe Lifts

Potchefstroom Pickup points

☒ Potchefstroom Ikageng

☐ Potchefstroom Promosa

☐ Potchefstroom Mohadin

Destination:

Johannesburg
Pretoria
Randfontein
Durban
Centurion
Bloemfontein
Capetown

☒ Return trip?

No. of Tickets:

Book lift

Enter correct number of tickets

OK

How to submit:

- Zip (compress) the assignment folder as per the instructions on eFundi.
- Submit the .zip as folder as per the instructions on efundi.